

eVTOL Patent Taxonomy & Embedding Structure — Progress Report

Prepared for: Phase Supervisor **Pipeline stage:** 11_taxonomy_structure_separation.ipynb (Batches 01 + 05, 1639-patent dataset), model facebook/dinov2-large (frozen), layers {18, 22, 24} × pooling {cls, mean_patch}

Section 1: Batches & Architecture Baseline Analysis

What this stage checks: whether the review spreadsheets and processed manifests agree end-to-end, and which DINOv2 layer/pooling configuration the rest of the report standardizes on.

Sourced from `Review_postprocess_Batch_01/05.xlsx`, `processing_manifest_Batch_01/05.csv`, and the notebook's Stage 0 outputs.

1.1 Individual and Combined Batch High-Level Summary

Table 1. Individual and combined batch high-level summary.

Metric	Batch_01	Batch_05	Global / Combined
Total patents reviewed (patent-level, T1)	352	211	563
Patents disapproved (isApproved=False)	60	87 ^❶	147
Patents approved (isApproved=True)	292	123	415
Total unique Patent IDs reviewed and approved into the architecture stage (T2) ^❷	125	95	220
Total architecture instances (approved, T2 “arch” rows)	301 ^❸	230 ^❸	531
is_main == True images	150	114	264

- ❶ One patent in Batch_05 has an unresolved review value; excluded from both counts (123+87+1=211).
- ❷ Lower than “approved” because approved-but-duplicate patents fold into an existing architecture record.
- ❸ Exceeds unique Patent IDs because one patent can disclose multiple aircraft configurations (14 multi-arch patents in Batch_01, 12 in Batch_05) — expected, not a data-quality issue.

Cross-check: 264 main images = 268 architectures – 4 with no approved main image. Reconciles exactly. `qc_issues.csv`.

Duplicate-Type Definitions

Table 2. Duplicate-type definitions and counts.

Type (as labelled in the data)	Batch_01	Batch_05	Plain-language meaning
“1 — Same Aircraft”	5	6	Same aircraft/design seen before; G1–M3 equal.
“2 — Images AND aircraft the same”	167	27	True exact duplicate; T2–M3 equal.
“3 — Same plane, small changes”	10	– (0)	Minor drawing variations; all taxonomy stages differ.

Type 2 (true exact duplicate) is the large majority (167/182 Batch_01, 27/33 Batch_05).

1.2 Global Batch Pipeline & Data Discrepancy Audit

Purpose: trace where the 563 reviewed patents narrow down to the 264 embedded images.

Table 3. Global batch pipeline & data discrepancy audit.

Pipeline Metric	Value
is_main == True images (manifest, both batches)	264
Total architectures labelled	268

Pipeline Metric	Value
Base patents behind those architectures	222
Taxonomy attributes (distinct fields) tracked	256
Images with a fully aligned embedding	264
Architectures missing an approved main image	4

Reconciling the Counts

268 vs. 256: different units, not a discrepancy. 268 = row count (one per architecture). 256 = column count (distinct taxonomy fields). Every row carries all 256 columns. `stage0/coverage_table.csv`.

268 vs. 264: all architecture main images are embedded; the 4 not embedded simply have no approved main image on disk. $268 - 4 = 264$, matching Section 1.1 exactly. `config.yaml`: `taxonomy.main_arch_only: false`; `qc_issues.csv`; `aligned_rows.csv`.

1.3 Optimal Layer and Patch Pooling Strategy Selection

Purpose: compare all 6 candidate DINOv2 layer × pooling configurations and decide which one the rest of the analysis standardizes on.

G1/M1/M2/M3 are taxonomy schema section codes (topology / fuselage-gear-boom / wing-empennage / propulsion), used downstream as labels against clusters. Baseline comparison across **DINOv2 layers × pooling**, on all 264 embedded images.

Note: only 3 of 5 columns are populated for all 6 candidates; silhouette/NMI were only run for layer 22 mean_patch (the bolded row), so this table doesn’t show a head-to-head winner — on Hopkins alone, layer 24 cls (0.724) scores higher. The bolded row is what was carried forward, decided by the criteria below the table, not by this table.

Table 4. Layer × pooling baseline comparison (all 264 embedded images).

Layer	Pooling	Effective dim (participation ratio)	Dims for 90% variance	Hopkins (clusterability)	Best clustering silhouette ⁴	Cluster → taxonomy alignment (best NMI) ⁴
18	cls	15.7	52	0.683	—	—
18	mean_patch	12.2	51	0.718	—	—
22	mean_patch	12.1	57	0.687	0.197 (k-means, k=2)	0.260 (M3_emp_chord)
22	cls	20.2	61	0.706	—	—
24	cls	26.0	61	0.724	—	—
24	mean_patch	12.1	57	0.664	—	—

⁴ Only computed for the reference matrix (layer 22, mean_patch); not run for the other 5. `config.yaml`: `taxonomy.reference_matrix`.

Selection Declaration: Layer 22, mean-patch pooling is used for all clustering/alignment work (Sections 3–4). Not derived from Table 4 — on Hopkins alone it’s the second-worst of six. Justified instead by the cosine-similarity QC histograms (Table 5): layer 22 mean_patch is the most balanced of the six (mean 0.940, std 0.029), avoiding two failure modes — **too collapsed** (layer 18, mean ≈0.94–0.98, little spread) and **too noisy** (layer 24 cls, mean 0.434, std 0.133, consistent with an unstable final-block [CLS] token). Section 4.3 then confirms it in practice: its best-aligning attribute (M3_emp_chord, NMI 0.260) beats every confound tested.

Scope — best available, not proven optimal. The spread argument cuts both ways: a more spread-out matrix (layer 24) could carry more signal if that spread is structure rather than noise, and that can’t be settled from histograms alone. The six matrices were never compared on taxonomy alignment directly — Section 4.4 does that, and in fact finds layer 22 cls ahead of layer 22 mean_patch on that criterion. **Treat this selection as provisional**, pending that follow-up comparison.

Set as taxonomy.reference_matrix in config.yaml.

Status: RECONCILED. **Core finding:** every count reconciles exactly (563 → 415 → 268 → 264); layer 22 mean_patch is the working reference matrix, chosen on cosine-spread balance, not proven optimality.

DATA AUDIT — RECONCILED

Section 2: Taxonomy Structure Sanity Check

What this stage checks: whether the embeddings are technically sound before drawing conclusions from them — no corrupted, duplicate, or dead vectors, no NaN/Inf, across all 6 matrices.

2.1 Embedding QC Matrix (Stage 1)

No corrupted/duplicate/dead embeddings found (N=264 figures):

Table 5. Embedding QC matrix — cosine-similarity statistics and integrity checks, per layer/pooling.

Layer	Pooling	Cosine mean	Cosine std	Cosine p05	Cosine p95	Exact duplicates	Near-dead dims	NaN/Inf
18	cls	0.9758	0.0118	0.9509	0.9896	0	0	0
18	mean_patch	0.9421	0.0348	0.8908	0.9783	0	0	0
22	cls	0.9029	0.0333	0.8369	0.9478	0	0	0
22	mean_patch	0.9398	0.0294	0.8905	0.9761	0	0	0
24	cls	0.4337	0.1332	0.2263	0.6666	0	0	0
24	mean_patch	0.8964	0.0417	0.8201	0.9541	0	0	0

Layer 24 cls is the outlier (mean cosine similarity drops to 0.43) — expected for the final block's [CLS] token, unstable late in ViT backbones.

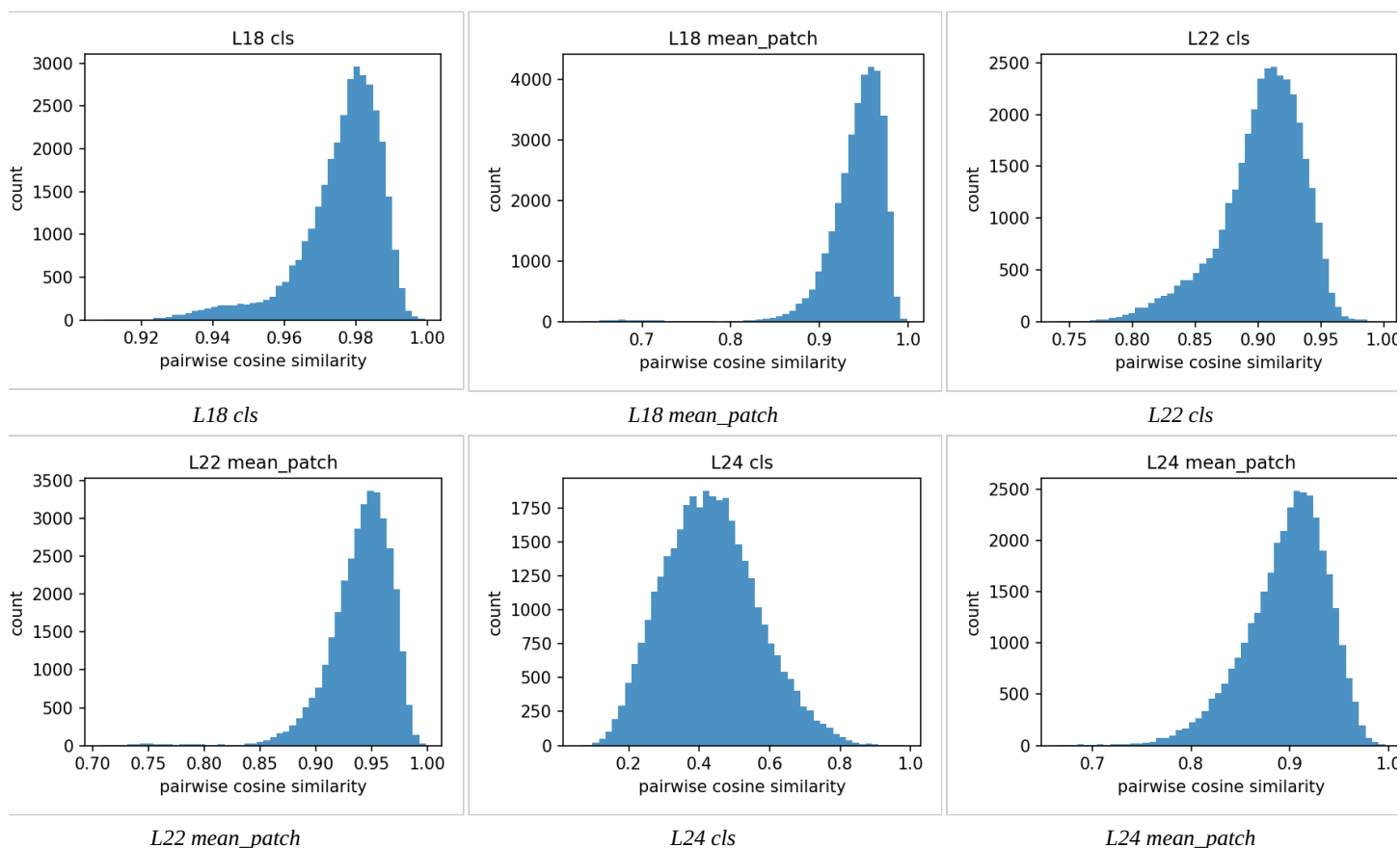


Figure 1. Cosine similarity histograms per matrix (regenerated as individual panels for legibility)

How to read: each panel is pairwise cosine similarity over 500 random pairs (1.0 = identical, 0 = unrelated). Pure QC, not a design-signal check — a histogram near 1.0 (layer 18) means little room for real design differences to show; a wider spread (layer 24 cls) means more room for structure, but also more noise-sensitivity. Whether the spread is real signal is for Sections 3–4 to test.

2.2 Sanity Check Validation

Status: PASSED. **Core finding:** all 6 matrices technically sound — zero duplicates/dead dims/NaN-Inf, distributions behave as expected per layer depth. Whether the near-1.0 concentration leaves enough *usable* separation is deferred to Sections 3–4.

Section 3: Global Embedding Structure vs. Random Noise

What this stage checks: whether the embeddings encode genuine geometric structure or are statistically indistinguishable from random noise — via PCA concentration and pairwise-distance vs. a random baseline.

3.1 PCA Structure Across Layers 18–24, cls vs. mean_patch

Table 6. PCA structure across layers 18–24, cls vs. mean_patch.

Layer	Pooling	PC1 variance ratio vs. random	Effective dim	Dims for 90% var
18	cls	20.43×	15.7	52
18	mean_patch	25.42×	12.2	51
22	cls	22.01×	20.2	61
22	mean_patch	26.57×	12.1	57
24	cls	18.27×	26.0	61
24	mean_patch	28.49×	12.1	57

mean_patch pooling shows a stronger dominant direction than cls, which spreads variance across more dimensions — mean_patch embeddings are more concentrated, cls more diffuse.

3.2 Pairwise Cosine Distance: Real Embeddings vs. Random Baseline

Random baseline = Gaussian noise of identical shape. All 6 matrices comfortably clear the pipeline’s $>3\times$ pass threshold.

Figure 2. PCA spectrum, real vs. random (per layer/pooling):

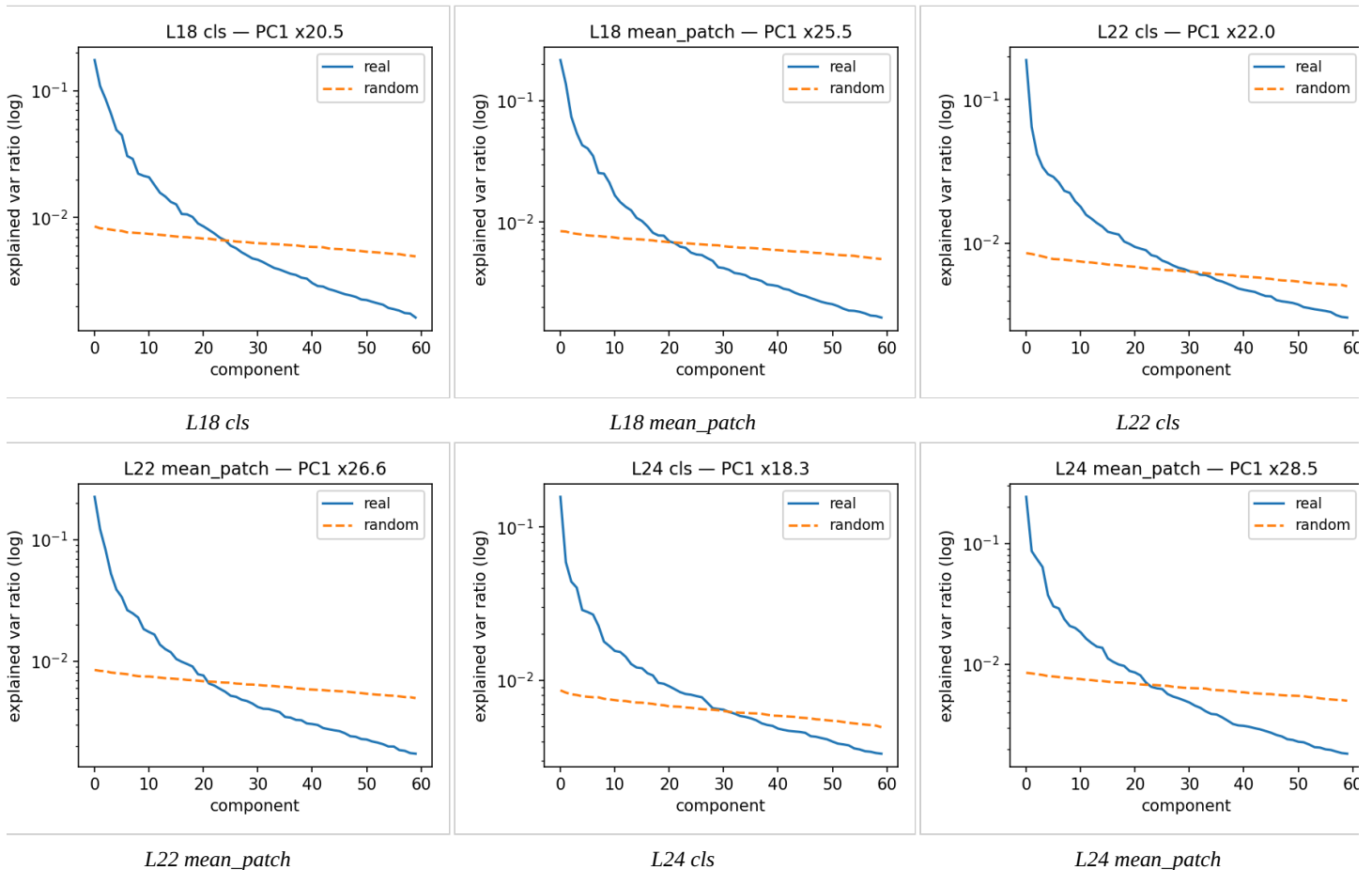
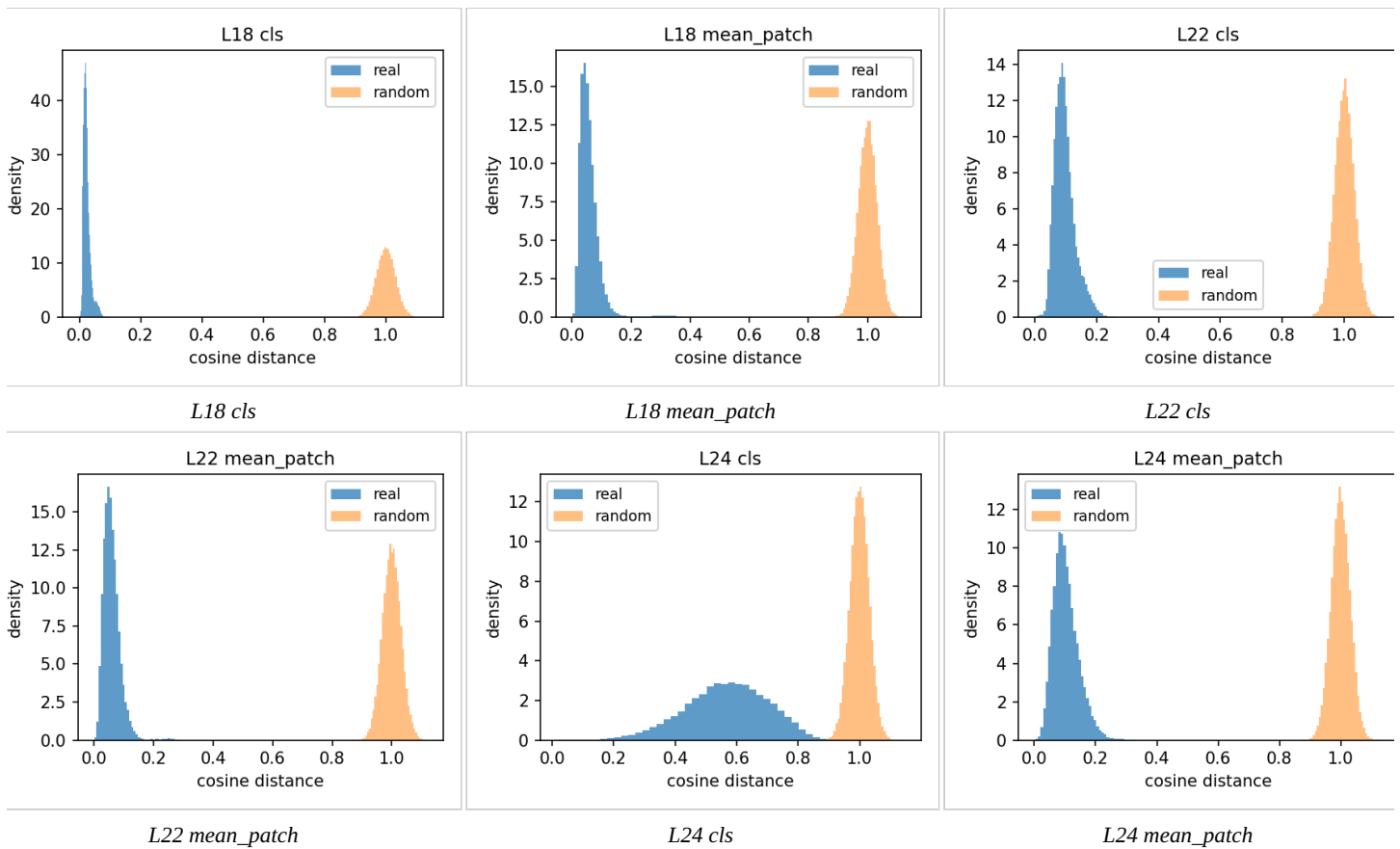


Figure 3. Pairwise cosine-distance distribution, real vs. random (per layer/pooling):



3.3 Structural Integrity Conclusion

All 6 matrices pass real-vs-random (PC1 18–28× baseline) and show non-trivial local structure (Hopkins ≈ 0.66 – 0.72 — a smooth continuum, not sharply separated islands). **Embeddings encode genuine geometric structure, not noise.** What that continuum is *made of* (design signal vs. drawing style) is answered downstream — Section 4.1 finds a branch grouped by rendering style, and Section 4.3 quantifies drawing perspective as a genuine confound (ARI 0.288).

Status: PASSED. **Core finding:** all 6 matrices clear the $\geq 3\times$ threshold by a wide margin, proving real geometric structure. Hopkins scores predict Section 4 will find coarse, gradient-like groupings rather than crisp clusters — which is what it finds.

REAL-VS-RANDOM TEST — PASSED

Section 4: Unsupervised Image-Level Clustering

What this stage checks: whether unsupervised clustering recovers meaningful aircraft-design groupings, or just superficial drafting/applicant style.

Using layer 22, mean_patch pooling — the pipeline’s configured reference matrix (Section 1.3).

4.1 Hierarchical Dendrogram

Purpose: visualize how the 264 architectures group hierarchically, before any cluster count is chosen.

Ward-linkage clustering over the 264-figure, layer-22-mean_patch matrix (PCA-57):

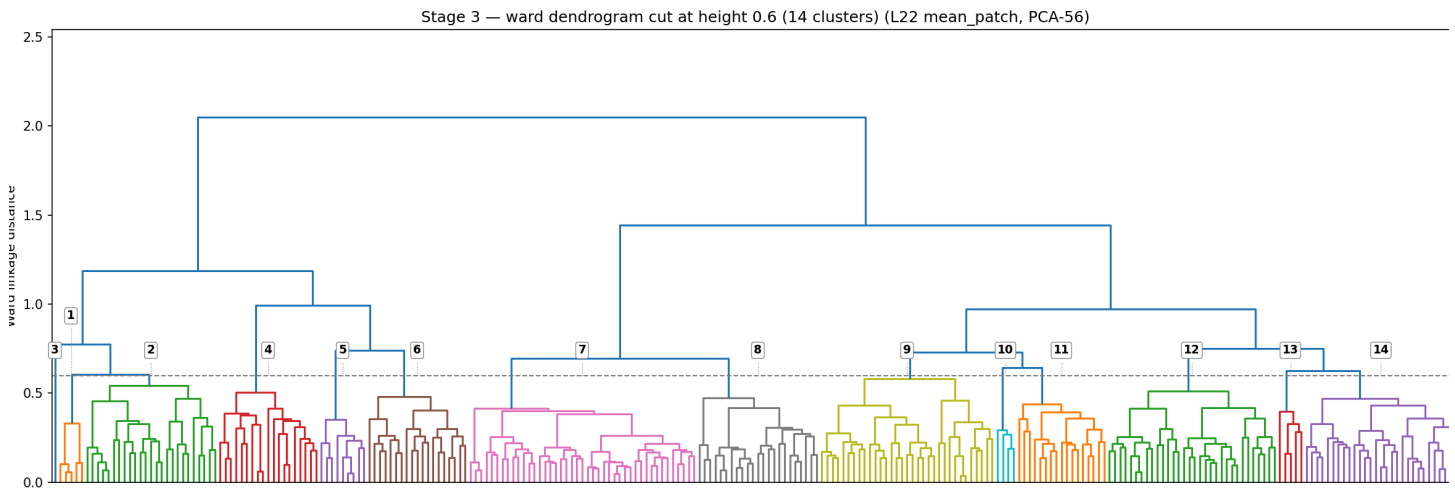


Figure 4. Dendrogram, labeled 1–14 at the cut used for the branch folders below

Branch-level image folders: cut at Ward distance 0.60 → exactly **14 branches** (a strict 0.47 cut gives 21 instead). Each labeled branch has a folder of its actual main images for auditing:

`docs/dendrogram_clusters/cluster_01/...cluster_14/` (1–43 images each; `cluster_sizes.json`).

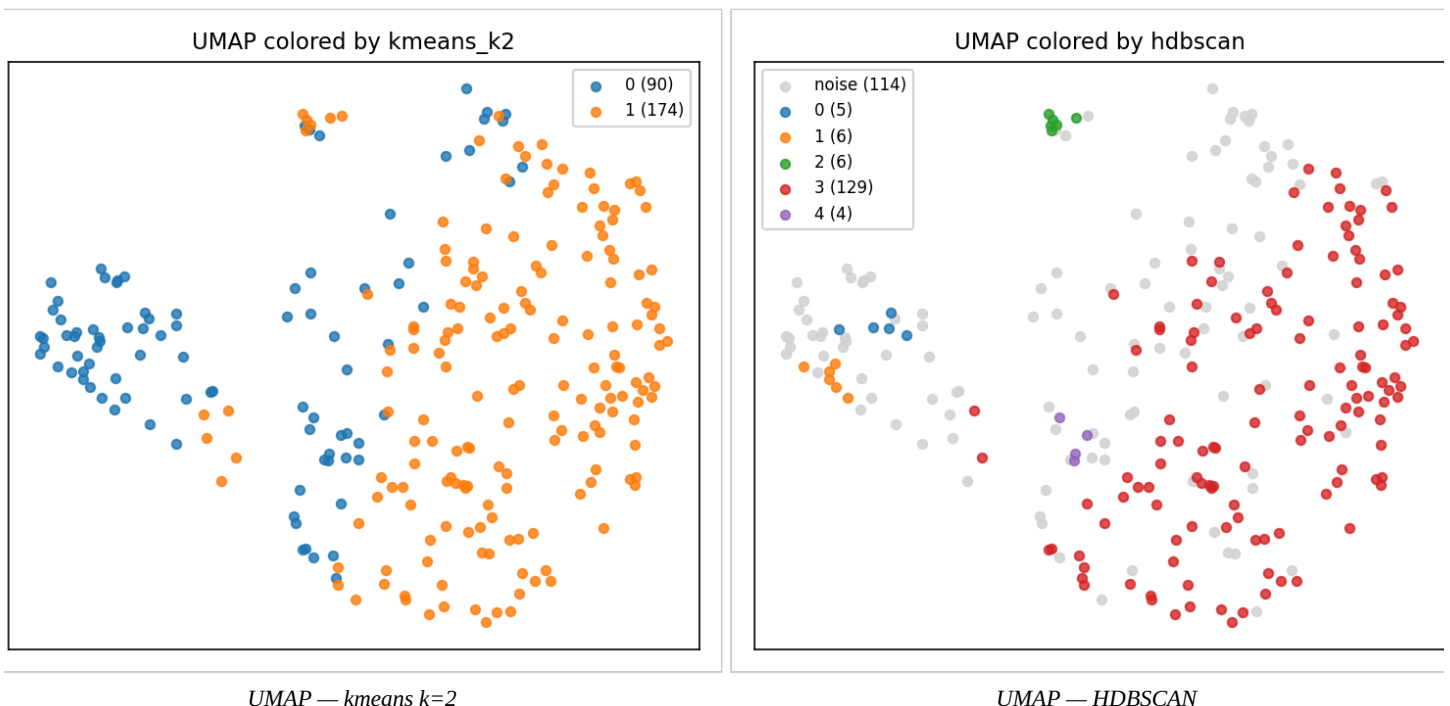
Visual spot-check: genuine consistency confirmed, but the common thread differs by branch — **branches 1 and 2** share drawing perspective more than design; **branch 7** shares a “cruise” configuration (similar fuselage/wing, mostly 2 tilting propulsors); **branch 14** is the same cruise style with more propulsors; **branch 13** shares rendering style (grayscale/shaded CAD) rather than design — the same effect as the drawing-perspective confound quantified in Section 4.3 (ARI 0.288).

4.2 Dimensionality Reduction Comparison

Purpose: find and validate the cluster count/algorithm that Sections 4.3–4.4 test against the taxonomy.

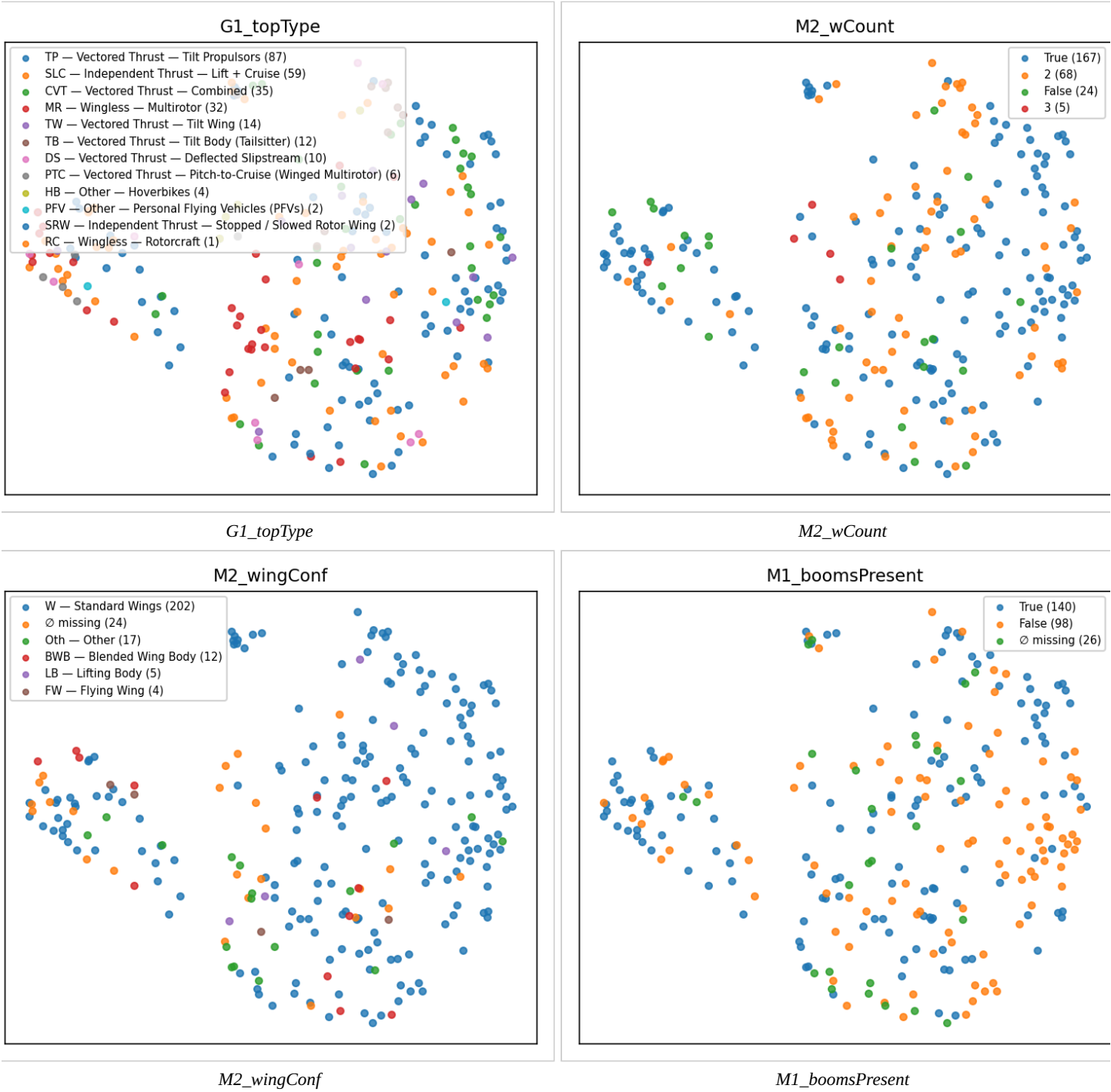
Conclusions up front: (1) a **coarse, stable 2-way split** — every algorithm agrees $k=2$ is best-supported (agglomerative silhouette 0.254, k-means 0.197, bootstrap ARI 0.724); (2) **finer structure exists but is fragile** — HDBSCAN finds 5 sub-groups at the cost of 43% “noise,” and $k=3$ –10 silhouettes all drop below 0.14; (3) taxonomy attributes appear as **gradients, not islands** in UMAP, matching Section 3’s “smooth continuum” verdict.

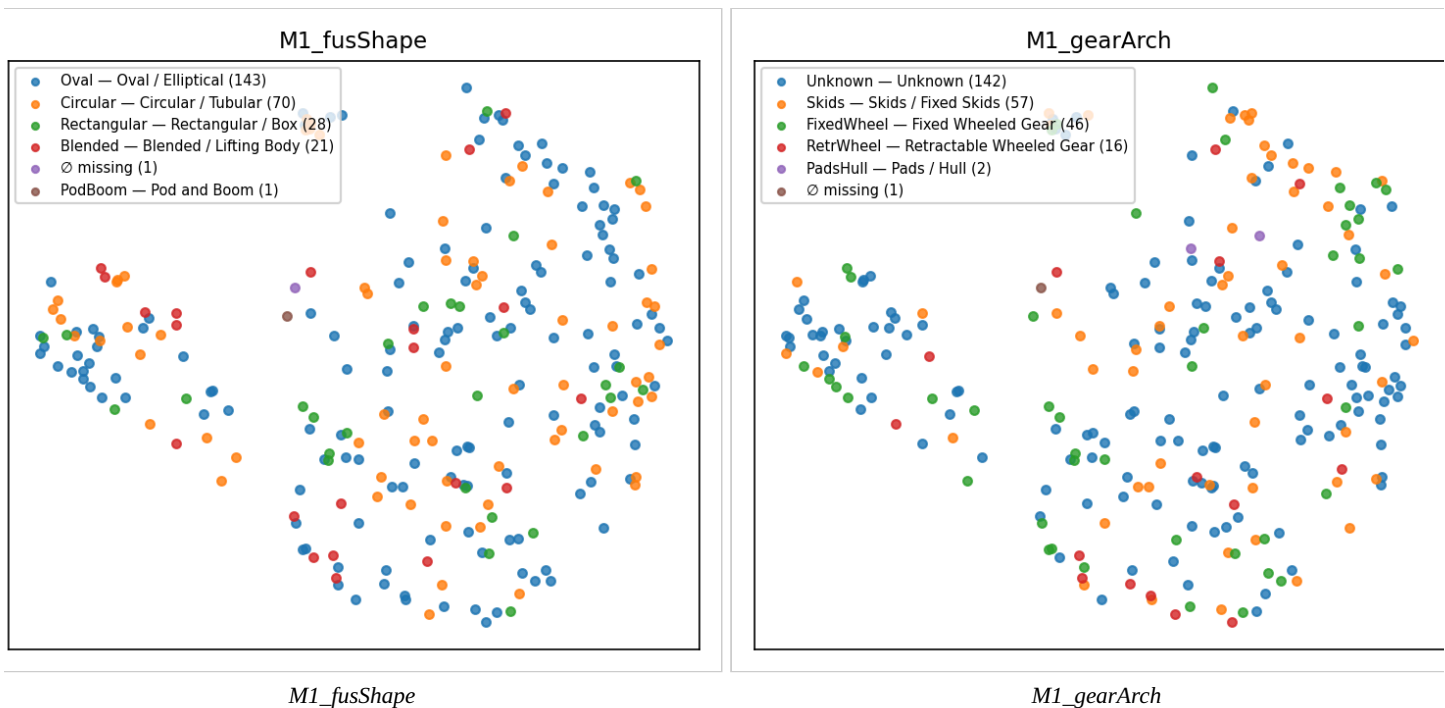
Figure 5. UMAP colored by unsupervised cluster (k-means $k=2$, HDBSCAN):



HDBSCAN’s largest cluster (129 images) occupies the same region as one k-means cluster; its “noise” points (114) fill the diffuse space between cores.

Figure 6. UMAP colored by taxonomy attribute (G1_topType, M2_wCount, M2_wingConf, M1_boomsPresent, M1_fusShape, M1_gearArch):





No single attribute cleanly reproduces the cluster boundary — hence the quantitative test in 4.3.

Clustering sweep: agglomerative k=2 (silhouette 0.254) best overall; k-means k=2 close behind (0.197); HDBSCAN settles on 5 clusters at 43.2% noise. Bootstrap stability (ARI) = 0.724 for k-means k=2.

4.3 Cluster ↔ Taxonomy Alignment (architectures)

Purpose: the key confound test — do clusters track real design attributes more than nuisance factors like applicant identity or drawing perspective?

NMI (0–1) measures how much an attribute reduces uncertainty about cluster membership, but is *not* chance-corrected — many small categories can inflate it artificially. **ARI** measures the same agreement but *is* chance-corrected (0 = random, can go negative). High NMI + comparable ARI = real signal; high NMI + near-zero ARI = artifact.

“**Categories / N**” = number of label values / number of images actually carrying that label (many fields only apply to a subset of aircraft). More categories over fewer labelled images drives NMI’s inflation problem.

Best-aligning attributes to the 2-cluster (k-means) partition, out of 79 tested:

Table 7. Best-aligning taxonomy attributes vs. confounds, 2-cluster (k-means) partition.

Attribute	NMI	ARI (chance-corrected)	Categories / N labelled	What it is
M3_emp_chord	0.260	0.218	2 / 39	Whether the empennage-mounted propulsor faces forward or back — puller (“Front”) vs. pusher (“Back”) configuration.
M3_emp_zone	0.235	0.123	5 / 41	Where on the empennage that propulsor sits — tip-mounted, stacked vertically, or stacked horizontally.
M3_boom_t2_count	0.145	−0.013	9 / 94	Number of secondary (type-2) propulsion units mounted on the boom, per boom.
M2_wing1_posV	0.139	0.210	4 / 211	Vertical mounting position of the main wing on the fuselage — high/shoulder, mid, or low.
M1_boom1_circSym	0.135	0.109	2 / 111	Whether the primary boom is circumferentially symmetric (round cross-section) or not.
<i>assignee (confound)</i>	0.207	0.036	109 / 264	Patent applicant/company — a drafting-style confound, not a design attribute.
<i>drawing perspective (confound)</i>	0.201	0.288	6 / 264	The figure’s viewpoint (front, side, top, isometric, ...) — reflects how the patent was drawn, not the aircraft’s design.

Attribute	NMI	ARI (chance-corrected)	Categories / N labelled	What it is
<i>assignee_country (confound)</i>	0.169	0.182	19 / 261	Applicant's country of origin — another drafting/institutional confound.

Cross-referencing NMI and ARI:

- **Assignee (mirage):** 109 companies / 264 patents inflates NMI to 0.207; ARI is 0.036 — real agreement is practically zero.
- **M3_boom_t2_count (false positive):** NMI 0.145 but ARI -0.013 — aligns worse than chance.
- **M3_emp_chord (real signal):** ARI (0.218) tracks NMI (0.260) closely — legitimate.
- **Drawing perspective (genuine confound):** ARI (0.288) is *higher* than its NMI (0.201) — a real, strong relationship, and the one most worth controlling for (see next steps).

4.4 Intra- vs. Inter-Class Distances (Notebook Stage 5 — layer comparison)

Purpose: per-attribute test of whether same-class patents sit closer together in embedding space, and which layer/pooling separates classes best.

Separation ratio = mean between-class cosine distance ÷ mean within-class distance (1.0 = attribute invisible to the embedding; 1.54 = different-class pairs 54% farther apart). Cohen's d is the standardized effect size; p-value checks it isn't a shuffling accident. Across 264 figures × 79 attributes (492 rows): **174 separate significantly at p<0.05** (excluding `cluster`).

cluster = the unsupervised k-means assignment fed back through this same test, as a reference ceiling — since clusters were fit to minimize exactly this distance, they should (and do) separate more strongly than any real attribute.

Top individual separations (excluding `cluster`):

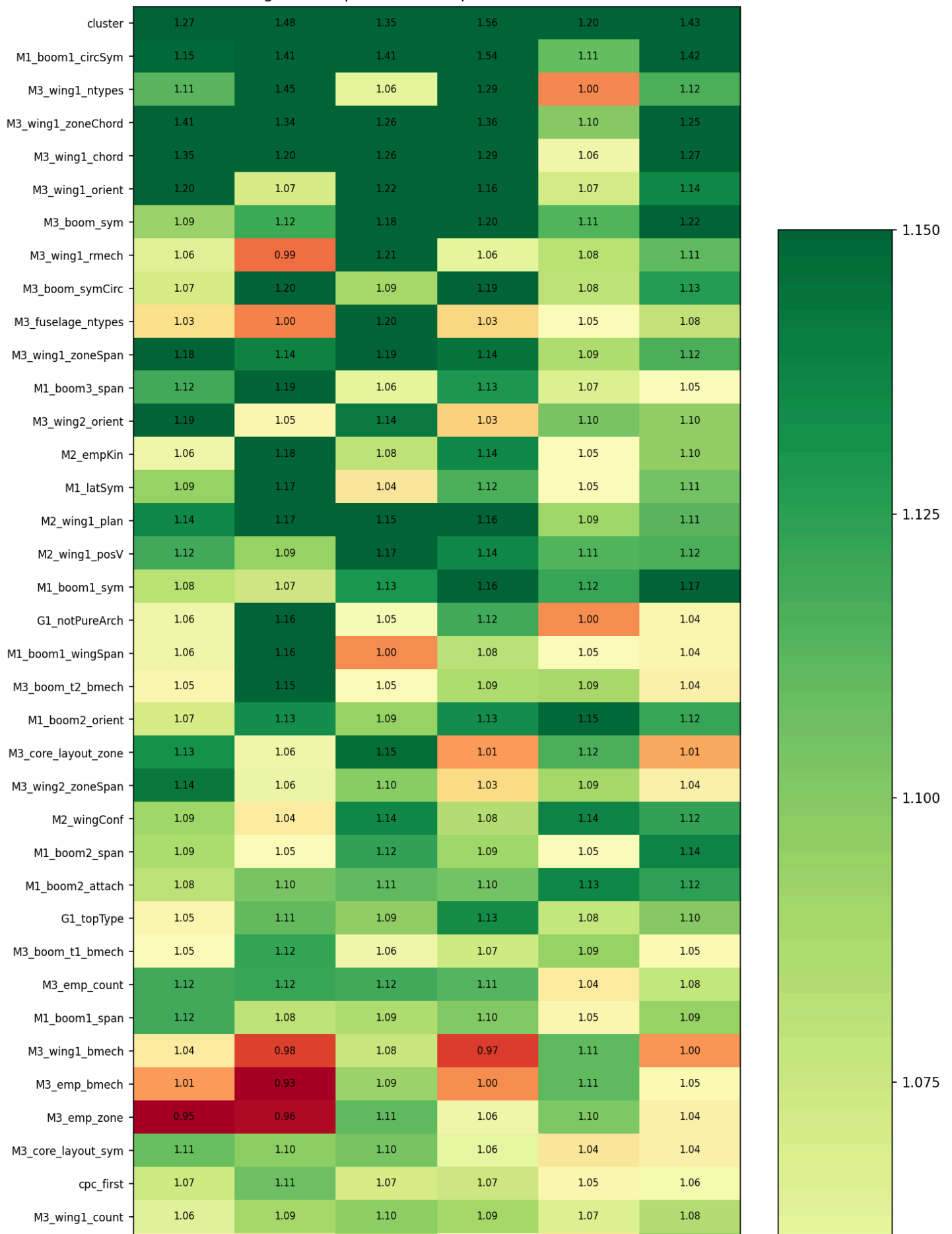
Table 8. Top individual attribute×layer separations (intra- vs. inter-class distance).

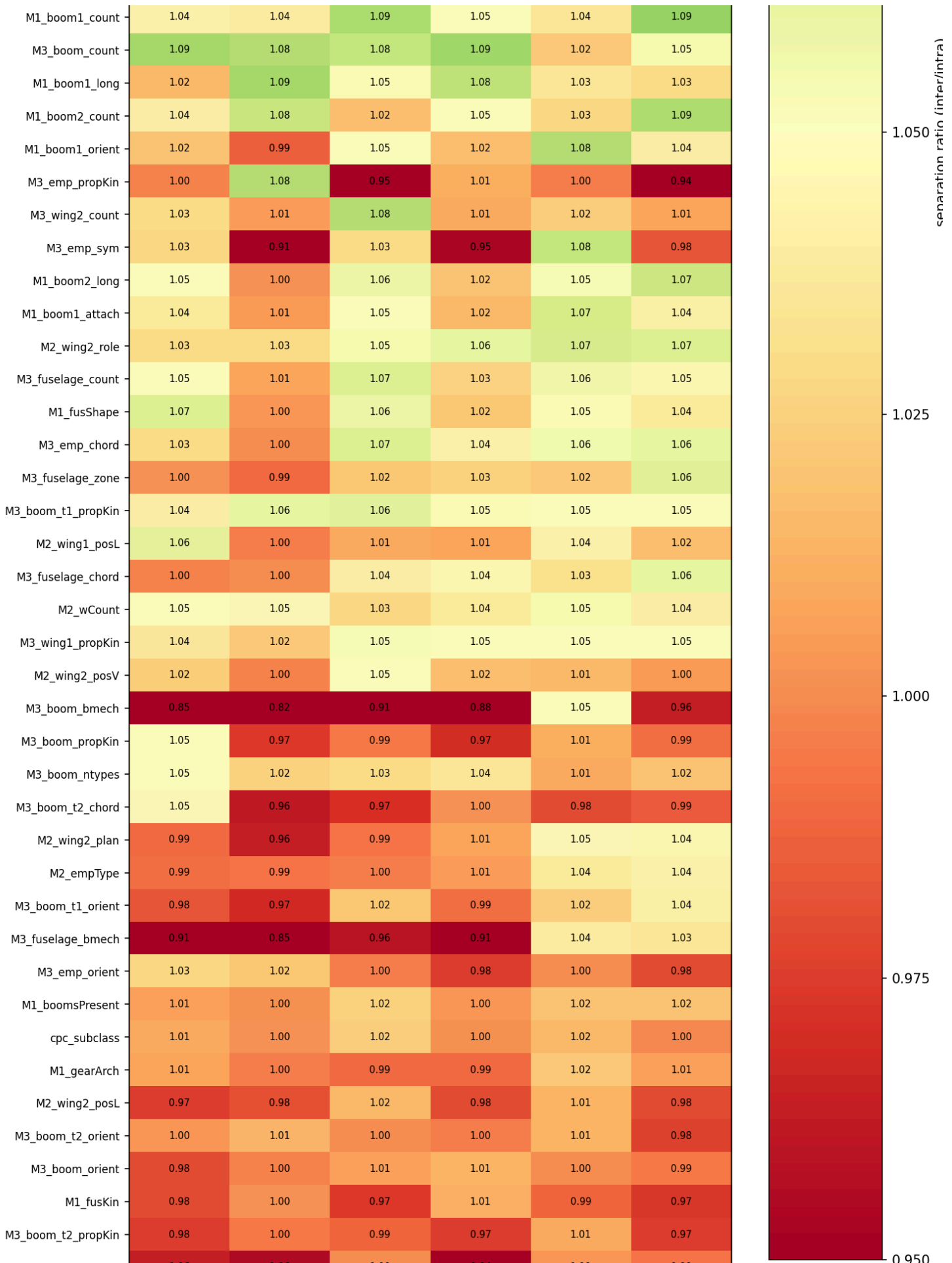
Attribute	Layer	Pooling	Separation ratio	Cohen's d	p-value
M1_boom1_circSym	22	mean_patch	1.543	1.248	0.001
M1_boom1_circSym	22	cls	1.412	1.149	0.001
M1_boom1_circSym	24	mean_patch	1.423	1.031	0.001
M1_boom1_circSym	18	mean_patch	1.415	0.911	0.004
M3_core_layout_zone	22	cls	1.146	0.735	0.005
M3_wing1_chord	22	cls	1.263	0.726	0.009

For reference, `cluster` itself separates most strongly (layer 22 mean_patch: ratio 1.560, d=1.03, p=0.001) — expected.

Averaging Cohen's d per layer×pooling, **layer 22 cls** has the highest effect size (mean d=0.363), ahead of layer 22 mean_patch (0.315) and the rest (0.26–0.30).

Stage 5 — separation ratio per attribute x matrix





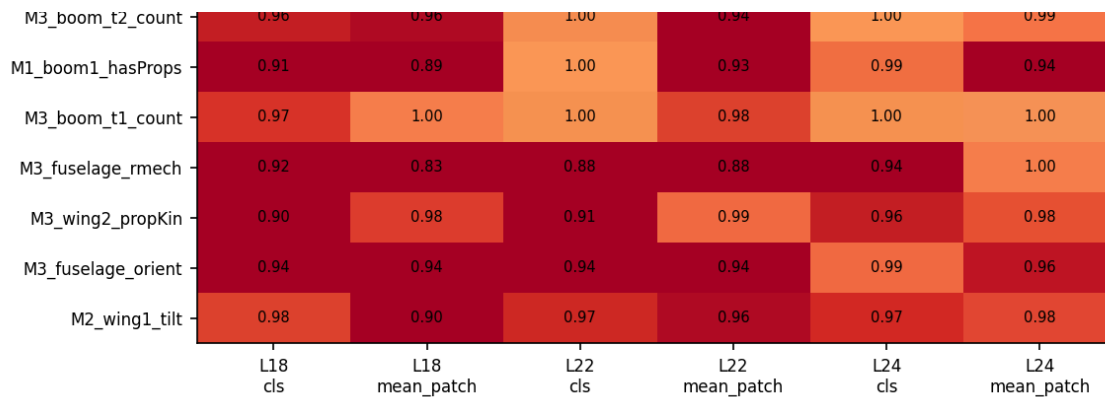


Figure 7. Separation heatmap

What this means for the reference matrix: this is the one test comparing all 6 matrices on a taxonomy-relevant criterion, and layer 22 **cls** edges out layer 22 **mean_patch** (0.363 vs. 0.315) — the layer-22 *depth* is validated, but the *pooling* choice is genuinely open. The margin is modest and this metric doesn't cover clustering quality or confound resistance, so rerunning Stages 3–4 on layer 22 **cls** (and layer 24 **mean_patch** as a secondary candidate) is listed as future work rather than done inline.

Status: PASSED, with a caveat. On NMI (the pipeline's actual gate), M3_emp_chord (0.260) beats every confound. Chance-corrected: M3_emp_chord is confirmed real (ARI 0.218), assignee dissolves (ARI 0.036), but drawing perspective is a genuine confound (ARI 0.288, above M3_emp_chord's own). **Core finding:** real design signal exists, but not yet *clean* — perspective must be explicitly controlled for before cluster membership is read as pure design signal.

CLUSTER ↔ TAXONOMY ALIGNMENT — PASSED (WITH PERSPECTIVE CAVEAT)

All tables/figures sourced from *DINov2_eVTOL_frozen_Analysis/notebooks/11_taxonomy_structure_separation.ipynb* and its outputs in *.../Preliminary_analysis/outputs/analysis_taxonomy/*.